

6. A dairy farm was using 15 000 kWh of electricity every month from the National Grid. The cost of electricity to the farm was 20 p per unit. The farmer decided to spend £152 400 on a biogas generator which uses cow dung to generate electricity. When working at maximum output, it was expected to reduce his demand on the National Grid to only 3 000 kWh a month.

(a) (i) Use an equation from page 2 to calculate the expected monthly savings in £. [3]

Savings = £

(ii) Calculate the payback time for the biogas generator. [1]

Payback time = months

(b) The biogas generator needs to produce 144 000 kWh of electricity per year. The farmer owns 150 cows. Each cow produces 200 kg of dung per week, but the farmer is only able to collect 60 kg of this. Each 1 kg of dung produces 0.05 m³ of methane gas. Each 1 m³ of methane gas input to the generator has an energy value of 5.3 kWh. Each 1 m³ of methane gas when burned produces an output of 1.9 kWh of electricity.

(i) Use the information above and an equation from page 2 to calculate the % efficiency of using methane gas to produce electricity. [2]

% efficiency =



Examiner
only

- (ii) The farmer thinks that there will be enough cow dung to produce the required amount of electricity (144 000 kWh).
Use the information opposite to explain whether the farmer is correct.
Show all your workings in the space below. (1 year = 52 weeks) [5]

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- (c) When cow dung decomposes it produces methane gas.
When methane gas burns, it produces the greenhouse gases carbon dioxide and water vapour.
An online article contains this information about methane:

“While carbon dioxide is said to be the major contributor to the greenhouse effect, methane is roughly 30 times more effective as a heat-trapping gas.”

Explain whether collecting cow dung to use in a biogas generator benefits efforts to reduce human impact on the greenhouse effect. [2]

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END OF PAPER

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